

Chapter 9 — IoT security, privacy, and overload

Always-on sensors create always-on risks

Learning objectives

- By the end of this part, learners should be able to explain scalability, security

Challenges: scalability, security, privacy

- As device counts grow
- Many IoT devices ship with minimal security, so each endpoint can be an attack surface
- Sensitive health and behavior data also demand compliance with privacy rules such as GDPR

Security practices for IoT and edge

- Best practices include end-to-end encryption at the device, in transit, and in the cloud
- These controls reduce unauthorized access, tampering

Example 9.20 — Securing Wearable Health Edge Devices

- Example 9.20 — hands-on module
- Example 9.20 highlights wearable health trackers that monitor vital signs in real time
- Securing those edge devices against tampering and ensuring health data reach clinicians
- Explore the chapter example module
- View files: ``modules/chapter9/example20/``

Example 9.21 — Smart-City Multi-Source IoT Collection

- Example 9.21 — hands-on module
- Public transit
- Protocols, and governance
- Explore the chapter example module
- View files: ``modules/chapter9/example21/``

Protocols and multi-source coordination

- Effective multi-source IoT depends on shared formats and lightweight protocols
- Message Queuing Telemetry Transport suits low-bandwidth messaging
- Standardization helps cities and operators fuse streams without brittle one-off

Ethics: ownership, overload, surveillance

- Who owns device-collected personal data, the individual or the vendor, is often contested
- Millions of devices also create overload: important signals can be lost in volume
- Public and personal tracking raise surveillance concerns

Takeaways

- IoT collection faces scalability, security
- Encryption, authentication, and secure edge devices are baseline controls
- Wearables and smart cities show why ethics, ownership, overload

Next

- Complete the quiz for this part
- The next part turns to big-data characteristics, volume, velocity, variety